

Computer Networking: Concepts

(CSE 3751)

Experiment 5

Aim:

Configuration and verification of **Network Address Translation (NAT)** on a router to allow communication between a private internal network and an external public network using Cisco Packet Tracer.

Objectives:

1. Understanding the concept of NAT used in Private Network.
2. Configuration of **Static NAT**, **Dynamic NAT** and **PAT (Port Address Translation)** in a router using both Config and CLI modes.
3. Verification of communication between internal and external networks using NAT.
4. Analysing the IP address translation through NAT table entries in a router during communication between Private internal network and external public network.

Exercises:

1. What is the purpose of NAT in IP networking? Why is NAT used in IPv4 networks?
2. What is Inside Local and Inside Global addresses?
3. What are the differences between **Static NAT**, **Dynamic NAT**, and **PAT**?
4. How does PAT allow multiple devices to share a single public IP address?
5. What command is used to view active NAT translations? How do you verify that NAT translations are working correctly in Cisco Packet Tracer?

AIM

Configuration and verification of Network Address Translation (NAT) on a router to allow communication between a private internal network and an external public network using Cisco Packet Tracer.

OBJECTIVES

- (1) Understanding the concept of NAT used in Private Networks.
- (2) Configuration of Static NAT, Dynamic NAT and PAT (Port Address Translation) in a router using both Config and CLI modes.
- (3) Verification of communication between internal and external networks using NAT.
- (4) Analysing the IP address translation through NAT table entries in a router during communication between Private internal network and external public network.

THEORY

• Introduction to NAT

Network Address Translation (NAT) is a method used on routers to translate private IPv4 address into a public IPv4 address for external communication.

• Need for NAT in Private Networks

Devices in private networks use private IPv4 addresses, which are non-routable on the internet, so NAT is required to allow communication with public networks.

• How NAT Works

When a device sends traffic outside the LAN, NAT replaces the private source IP with the router's public IP address, enabling the packet to reach external servers.

• Benefits of Using NAT

NAT helps conserve public IPv4 addresses, hides internal network structure, and allows multiple internal hosts to share a single public IP.

• Role of NAT in Secure Communication

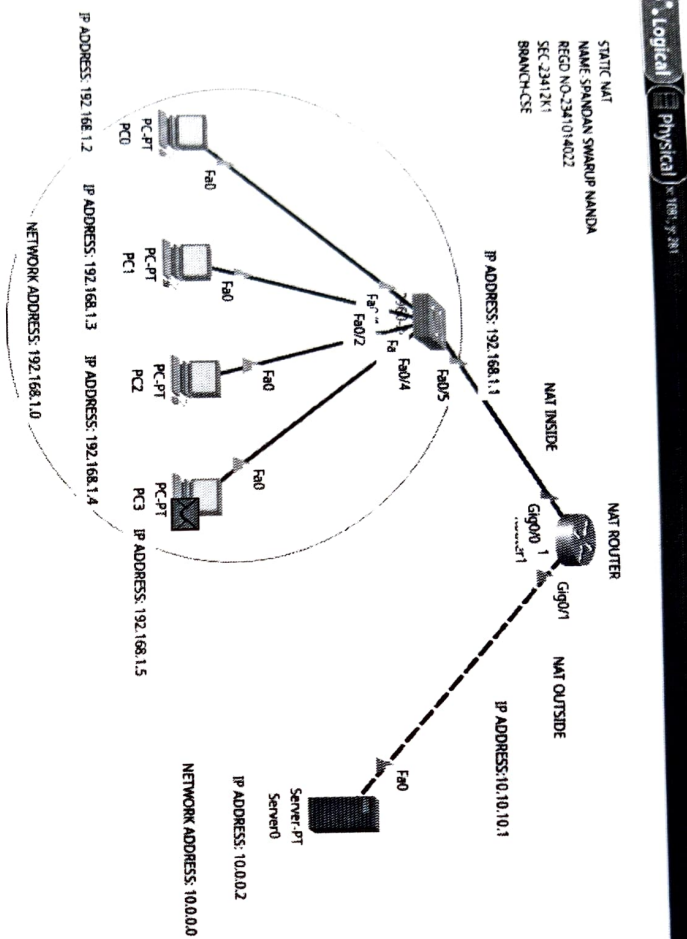
By masking internal addresses, NAT adds a layer of basic security, preventing direct access to internal devices from the internet.

P.T.O.

OBSERVATIONS

(i) Static NAT

Fig. Static NAT Configuration: Mapping local IPs to Global IPs.
This setup demonstrates Static Network Address Translation (NAT), where specific private IP addresses (like 192.168.1.2) are manually mapped one-to-one to public IP addresses (like 10.0.0.3). The CLI output confirms the translation table, allowing internal hosts to access external networks (the Internet) using a consistent public identity.



```
Router#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
---  ---
... 10.0.0.3            192.168.1.2       ...
... 10.0.0.4            192.168.1.3       ...
... 10.0.0.5            192.168.1.4       ...
... 10.0.0.6            192.168.1.5       ...
```

Router#
R1S-5-CONFIG-1: Configured from console by console

Copy

Paste

(ii) Dynamic NAT

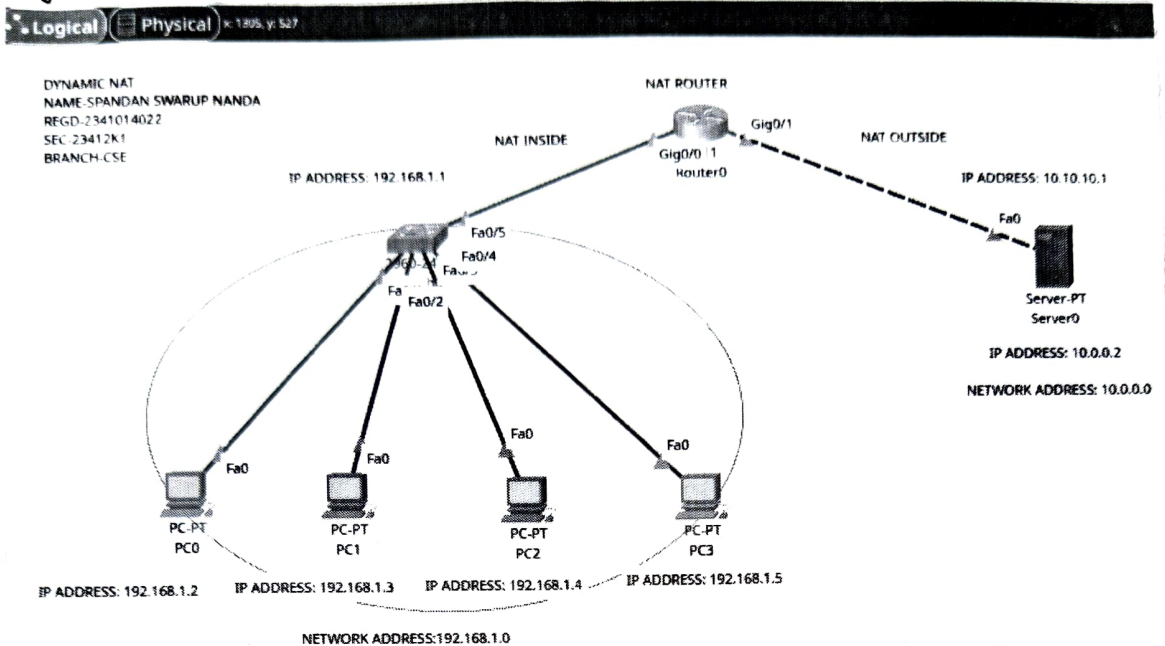


Fig. Dynamic NAT Topology : On demand IP Translation.

The figure illustrates a Dynamic NAT setup, where internal hosts (192.168.1.x) are assigned a public IP address from a defined pool only when they initiate a connection to the outside server.

Fig. Dynamic NAT CLI Configuration.
The figure shows the successful configuration of Dynamic NAT. By creating a public IP pool (NATPOOL) and an Access Control List (ACL) to identify internal traffic, then linked them together, so that private IPs are dynamically translated to public IPs.

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 10.10.10.1 255.0.0.0
Router(config-if)#ip address 10.10.10.1 255.0.0.0
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#interface gig 0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#interface gig 0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat pool NATPOOL 10.0.0.15 10.0.0.25 netmask 255.0.0.0
Router(config)#ip inside source list 1 pool NATPOOL
Router(config)#
% Invalid input detected at 'A' marker.

Router(config)#ip nat inside source list 1 pool NATPOOL
Router(config)#exit
Router#
%SYS-5-CONFIG-I: Configured from console by console
```



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router(config-if)#interface gig 0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#interface gig 0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat pool NATPOOL 10.0.0.15 10.0.0.25 netmask 255.0.0.0
Router(config)#ip inside source list 1 pool NATPOOL
^
% Invalid input detected at '^' marker.

Router(config)#ip nat inside source list 1 pool NATPOOL
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip nat translation
Router#show ip nat translations
Router#show ip nat translations
Pro Inside global Inside local Outside local Outside global
icmp 10.0.0.15:1 192.168.1.2:1 10.0.0.2:1 10.0.0.2:1
icmp 10.0.0.15:2 192.168.1.2:2 10.0.0.2:2 10.0.0.2:2
icmp 10.0.0.15:3 192.168.1.2:3 10.0.0.2:3 10.0.0.2:3
icmp 10.0.0.15:4 192.168.1.2:4 10.0.0.2:4 10.0.0.2:4
icmp 10.0.0.16:2 192.168.1.3:2 10.0.0.2:2 10.0.0.2:2
icmp 10.0.0.16:3 192.168.1.3:3 10.0.0.2:3 10.0.0.2:3
icmp 10.0.0.16:4 192.168.1.3:4 10.0.0.2:4 10.0.0.2:4
icmp 10.0.0.16:5 192.168.1.3:5 10.0.0.2:5 10.0.0.2:5
icmp 10.0.0.17:2 192.168.1.4:2 10.0.0.2:2 10.0.0.2:2
icmp 10.0.0.17:3 192.168.1.4:3 10.0.0.2:3 10.0.0.2:3
icmp 10.0.0.17:4 192.168.1.4:4 10.0.0.2:4 10.0.0.2:4
icmp 10.0.0.17:5 192.168.1.4:5 10.0.0.2:5 10.0.0.2:5
icmp 10.0.0.18:3 192.168.1.5:3 10.0.0.2:3 10.0.0.2:3
icmp 10.0.0.18:4 192.168.1.5:4 10.0.0.2:4 10.0.0.2:4
icmp 10.0.0.18:5 192.168.1.5:5 10.0.0.2:5 10.0.0.2:5
icmp 10.0.0.18:6 192.168.1.5:6 10.0.0.2:6 10.0.0.2:6

Router#
```

Fig. Configuring Dynamic NAT via CLI

This sequence shows the essential commands to enable Dynamic NAT. The configuration defines the "inside" and "outside" interfaces, creates a pool of public IPs (NATPOOL), and uses an Access List (ACL 1) to authorize the internal network to use that pool.

ANALYSIS

1. Static NAT Configuration Analysis

In this setup, multiple inside hosts from the 192.168.1.0/24 private network were mapped to unique public IP addresses on the 10.0.0.0/24 network using static NAT rules. The router interfaces were correctly configured as:

- G0/0 → NAT Inside (192.168.1.1)
- G0/1 → NAT Outside (10.10.10.2)

Static NAT entries were added in the CLI using commands such as:
ip nat inside source static 192.168.1.2 10.0.0.3

These commands created a permanent one-to-one mapping between each inside local address and a fixed address.

The command:

show ip nat translations

displayed all static mappings exactly as configured, confirming that NAT was functioning properly. Static NAT ensures each PC maintains a consistent public identity, which is required for services on devices that must always be reachable from outside networks.

2. Dynamic NAT Configuration Analysis

In the dynamic NAT setup, the inside network 192.168.1.0/24 was translated using a pool of public addresses from 10.0.0.15 to 10.0.0.25. The router interfaces were again configured as inside and outside, and an ACL was used to define which internal addresses could be translated.

CLI configuration included:

```
access-list 2 permit 192.168.1.0 0.0.0.255  
ip nat pool NATPOOL 10.0.0.15 10.0.0.25 netmask 255.0.0.0  
ip nat inside source list 1 pool NATPOOL
```

This set up allowed first-come, first-served translation, where inside hosts were dynamically assigned available public IPs from the NAT pool.

The output of:

show ip nat translations

displayed multiple dynamic entries showing inside local addresses mapped to different inside global address from the pool. These mappings were created only when traffic was initiated, proving that dynamic NAT was operating correctly. Dynamic NAT provides flexibility and efficient utilization of public addresses, as clients do not require fixed mappings and addresses are reused as connections close.